



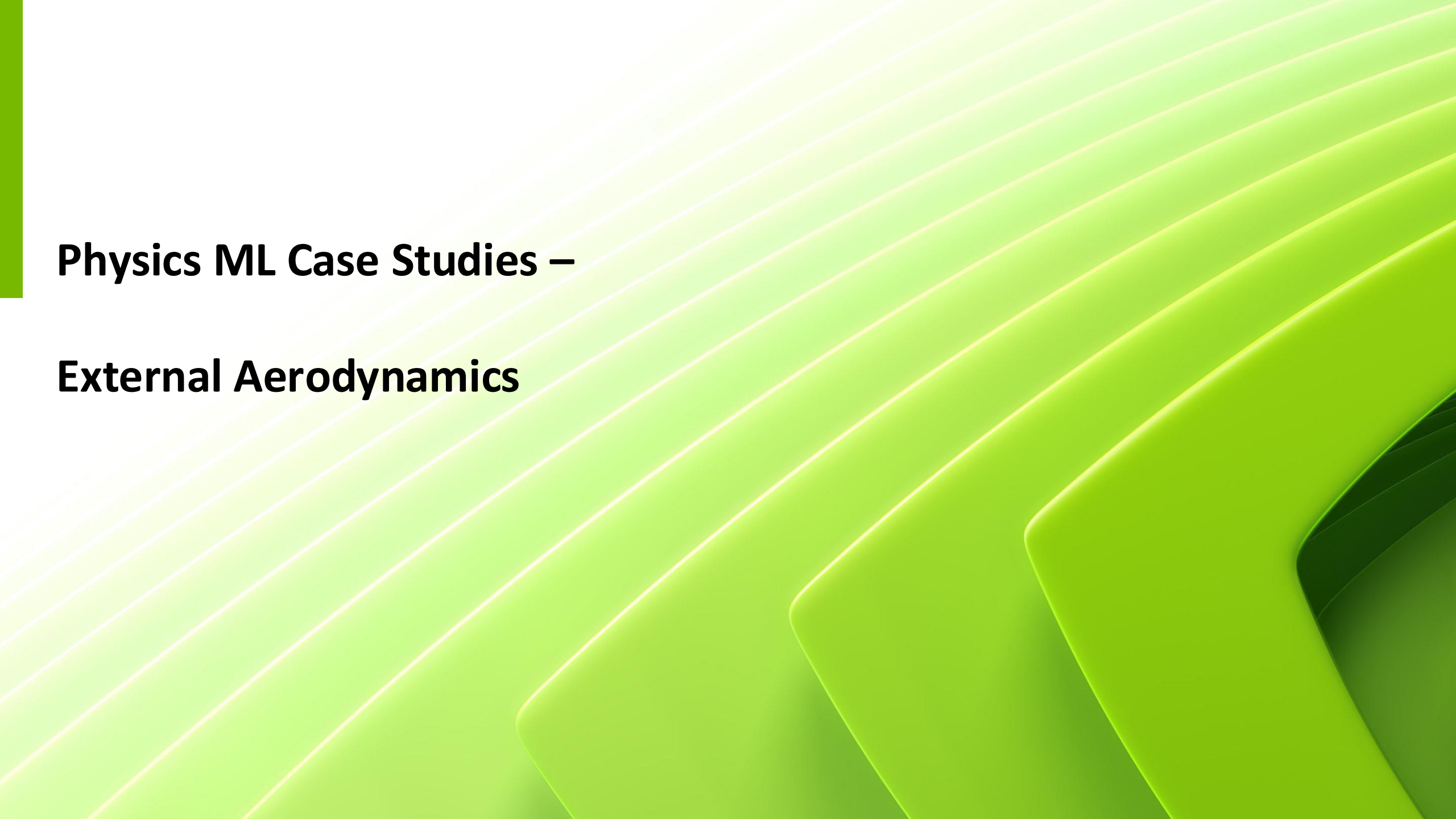
國家實驗研究院
國家高速網路與計算中心
National Center for High-performance Computing

NVIDIA PhysicsNeMo for NCHC AI-Powered Physics Bootcamp

Chao-Shun Zhan | Solutions Architect, Mar. 2026

Agenda

- Deep Dive – Case Studies
 - External Aerodynamics
- Hands-on with PhysicsNeMo — Challenges



Physics ML Case Studies –

External Aerodynamics

CFD simulations

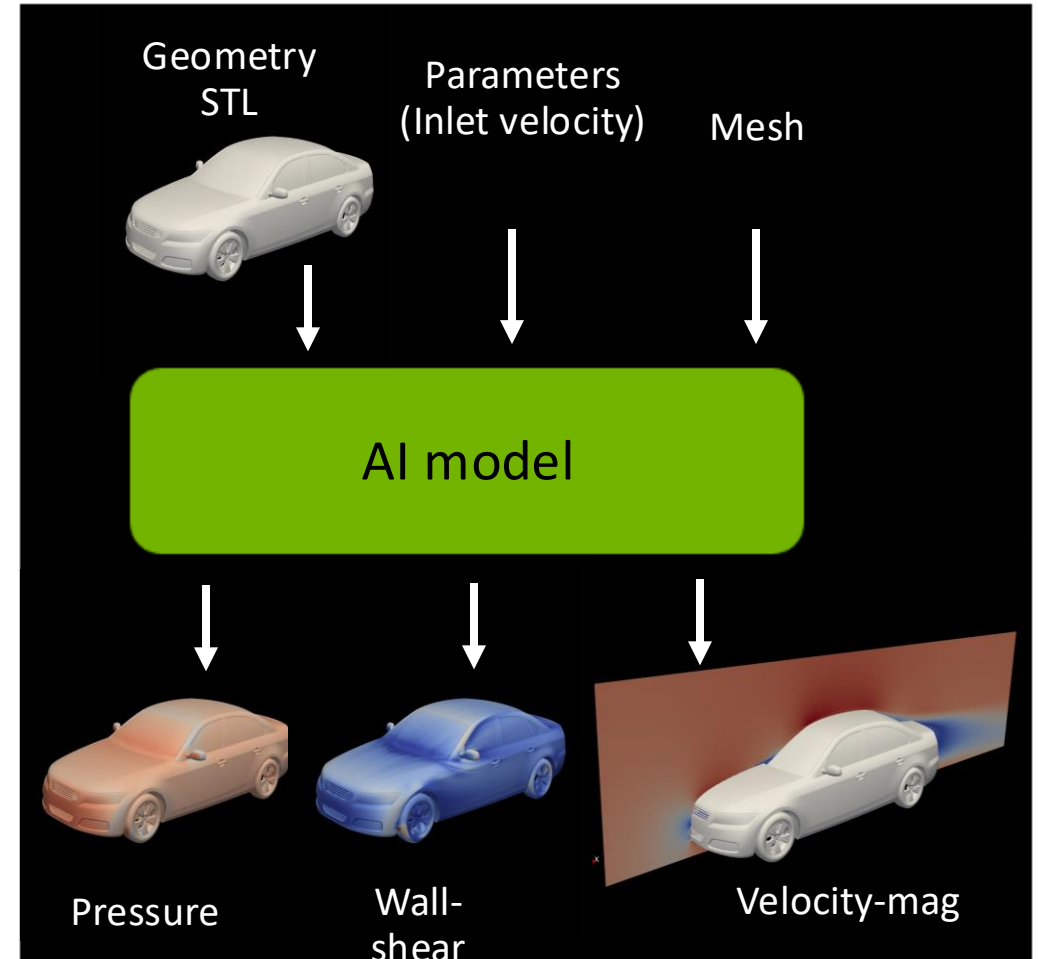
Automotive External Aerodynamics

- Automotive External Aerodynamics - predict the flow over automotive geometries
- Traditional finite difference simulations are expensive.
 - Solving Navier Stokes equations on high resolution domain of the order of 100 million mesh cells
- Develop AI surrogate model(s) – predict velocity, pressure, turbulent kinetic energy, wall shear stress, drag force for any geometry (STL of any roadside vehicle, free stream velocity).



Challenges: AI for Automotive Aerodynamics

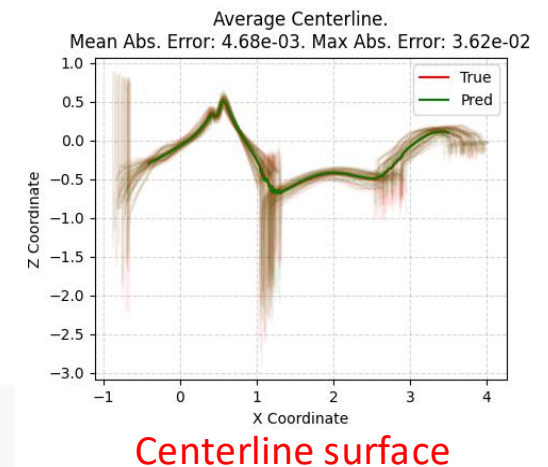
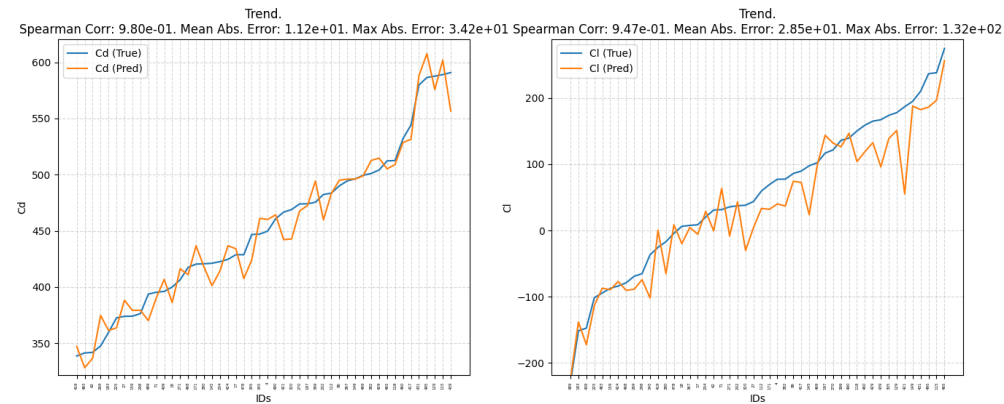
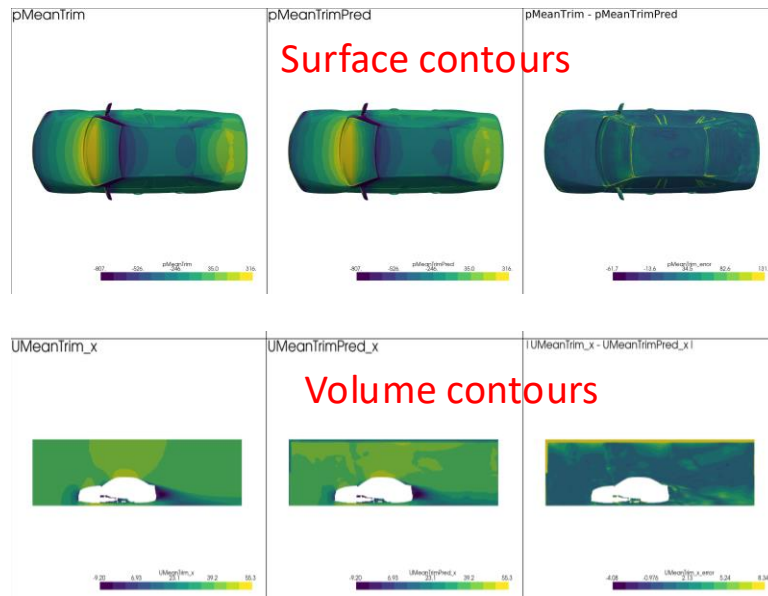
- Scalability to large simulation meshes
 - Auto-aero simulations have volume meshes of the order 100s million elements and surface meshes of 10s million elements
 - Tradeoff between scale and accuracy/generalizability, scaling AI models to high-res meshes impacts accuracy
 - Need optimized datapipes and post-processing tools for supporting surrogate models
- Model architectures
 - AI models are focused on learning global representations
 - Impacts ability to capture long-range interactions and fine-scale geometry features
 - AI models are dependent on spatial structure of mesh/point cloud, Don't generalize to arbitrary geometry or mesh representations



External Aerodynamics benchmarking

Utilities and workflows for benchmarking ML models (coming soon)

- Developed utilities and workflows for benchmarking to analyze ML models in a consistent and transparent manner using CAE specific metrics
 - Users train their ML models with the OSS DrivAerML dataset (<https://caemldatasets.org/drivaerml/>) with a specified train and test split
 - Trained models are used to predict on test set and written back into the VTPs and VTUs provided with the dataset
 - Use our benchmarking utilities with these files to generate CAE specific results (few examples showed below)
 - L_2 error metrics, drag force R^2 coefficients and design trends, comparisons of surface and volume contours, centerline plot on surface, line plots in different volume regions such as wake, underbody etc. for different field variables



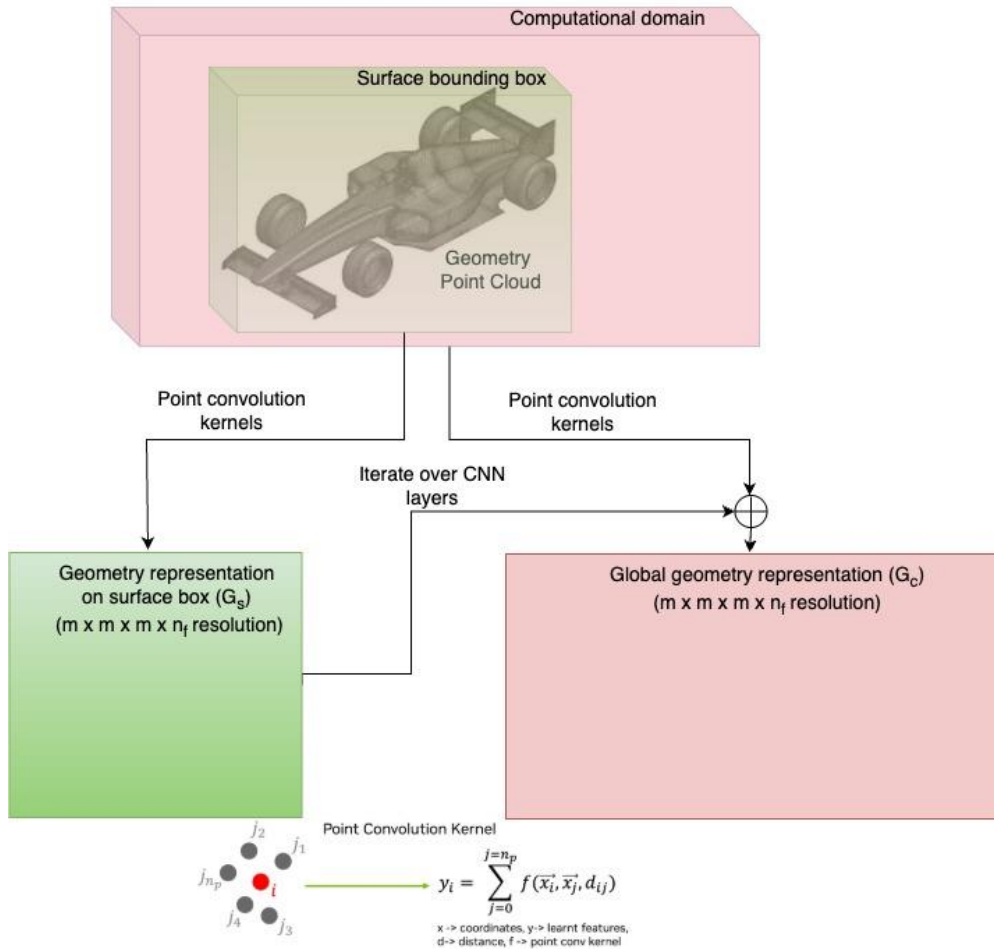
DoMINO

Point cloud based neural operator for scalability, accuracy and generalizability

- DoMINO stands for Decomposable Multiscale Iterative Neural Operator
- Key features:
 - **Neural operator:** Predicts point-wise volume and surface fields, infinitely scalable
 - **Decomposable:** Learns local geometry representations in sub-regions to improve solution accuracy
 - **Multi-scale:** Learns multi-scale point kernels to capture fine- and coarse-scale geometry features
 - **Iterative:** Facilitates long-range interaction by propagating geometry features into computational domain
 - **Basis in traditional numerical methods:** Builds dynamic computational stencils to learn non-linear basis functions
 - **Only STLs required at inference (no surface or volume meshes):** Insensitive to spatial structure and density of point cloud at inference
- DoMINO artifacts:
 - Preprint <https://arxiv.org/pdf/2501.13350>
 - Source code: <https://github.com/NVIDIA/physicsnemo/blob/domino/physicsnemo/models/domino/model.py>

DoMINO model architecture

Geometry NN and Aggregation NN

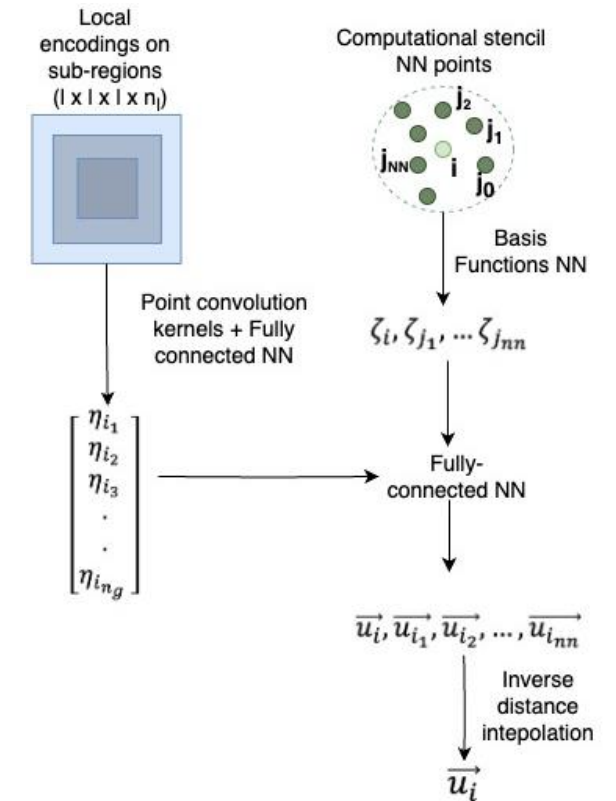


Geometry NN:

- Projects STLs on structured representations using dynamic point convolution kernels
- Propagates encoded features from surface to computational domain using iterative CNN layers

Aggregation NN:

- Extracts local geometry encodings in multi-res subdomains from global representations
- Dynamically constructs a finite-volume stencil and approximates solution on the cell center conditioned on the local geometry representation



Validation studies

DoMINO model validated on 3 use cases:

- DrivAerML (<https://arxiv.org/pdf/2408.11969>):
 - Publicly available, high-fidelity hybrid LES/RANS dataset comprising of volume and surface data for 500 geometrically morphed variants of the DrivAer Notchback car
 - ~150 mil volume and ~10 mil surface meshes
 - 430 used for training and 50 for testing
- DriveSimCFD-1k:
 - Internal steady-state RANS dataset generated using OpenFOAM. Comprises of 1000 geometrically morphed variants of Sedans, SUVs, Hatchbacks, Pickups, Vans etc. simulated at speeds ranging from 20 to 50 m/s
 - ~50 mil volume and ~5 mil surface meshes
 - Tested on 40 unseen cars and speeds and additional 30 DrivAerML models
- SHIFT-SUV dataset by Luminary Cloud (<http://shift.luminarycloud.com/>)

NVIDIA DoMINO-Automotive-Aero NIM

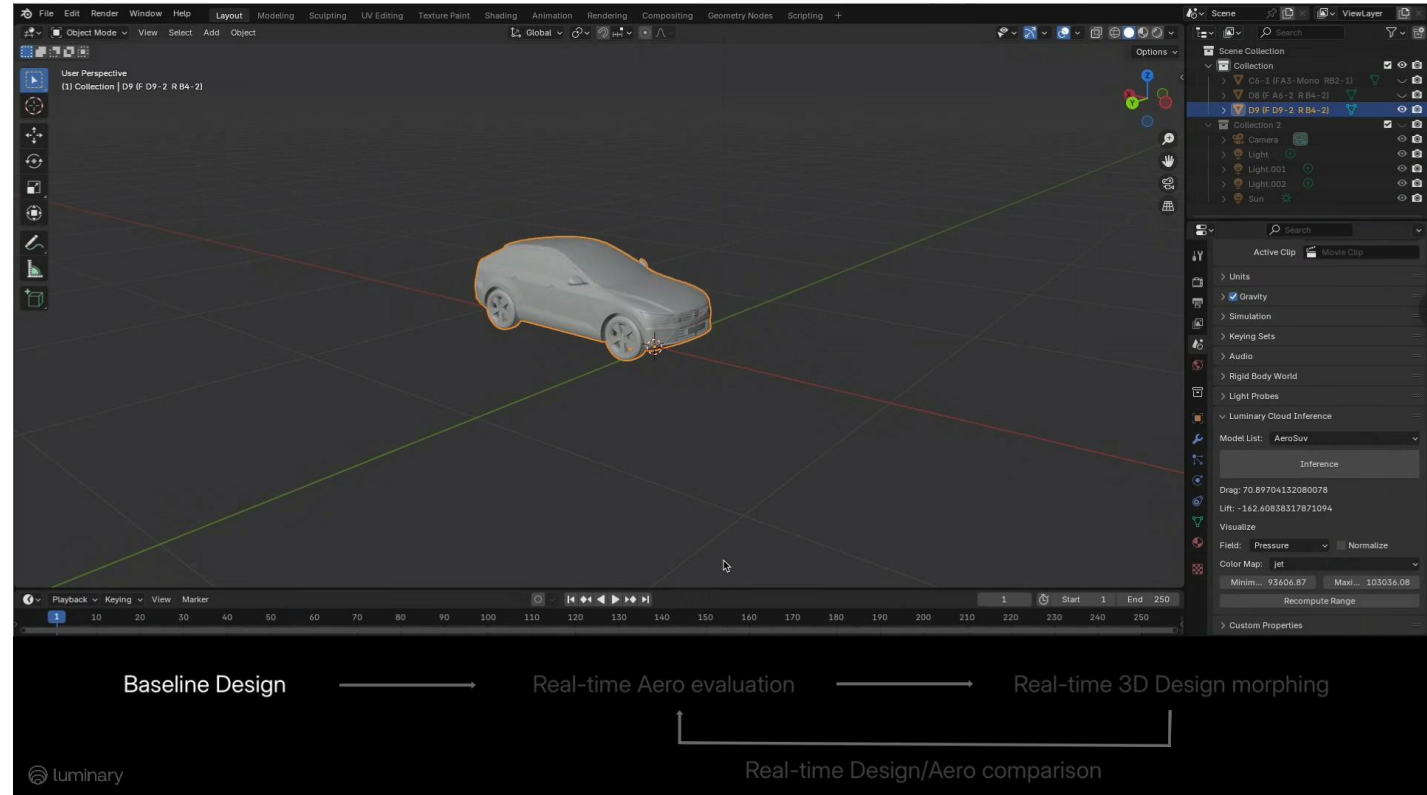
- **Pretrained DoMINO for road vehicle aerodynamics**
 - Trained on dataset spanning across 5 classes of cars (Sedans, SUVs, Pickups, Hatchbacks and Vans) across flow regimes (35-145 mph)
 - Predicts surface and volume flow fields, and aerodynamic forces from vehicle STLs and inflow speeds
 - Starting point for early-stage design evaluation and finetuning to custom datasets
 - Enables hybrid workflows (e.g., solver initialization, mesh guidance)
- **For use by engineering ISVs**
 - ISVs can accelerate model development and enable hybrid solver workflows
 - NIM can be further finetuned on custom datasets
- **NIM stands for NVIDIA Inference Microservices.**
 - Pre-trained AI model checkpoint with optimized inference runtime
 - Packaged as a container with well defined API
 - Deployable on-prem and in-cloud



Luminary Cloud

Physics AI CFD solution using PhysicsNeMo in 2 months

- Luminary uses PhysicsNeMo and the DoMINO architecture for their Physics AI solution
- Luminary in collaboration with nTop were able to develop a rapidly iterative design optimization workflow for their customers.



<https://www.luminarycloud.com/blog/introducing-luminary-shift-models-a-suite-of-physics-ai-foundation-models-to-transform-engineering-design/>

Demo

Digital Twin for Interactive Fluid Simulation

- <https://build.nvidia.com/nvidia/digital-twins-for-fluid-simulation>

Input

Camera View ⓘ
Side View ▾

Vehicle ⓘ
Model 1 ▾

Rim Type ⓘ
Aerodynamic ▾

Features ⓘ
☒ Mirrors ☒ Spoilers

Ride Height ⓘ
☐ Lowered ☐ Standard ☒ Lifted

GOVERNING TERMS: The trial service is governed by the [NVIDIA API Trial Terms of Service](#); use of the model is governed by the [NVIDIA AI Foundation Models Community License Agreement](#).

Output

Curve Trace Volume Trace Volume Slice

Speed Control ⓘ
0 50 100

Velocity Magnitude ⓘ
0 50 100

0 46 92 140



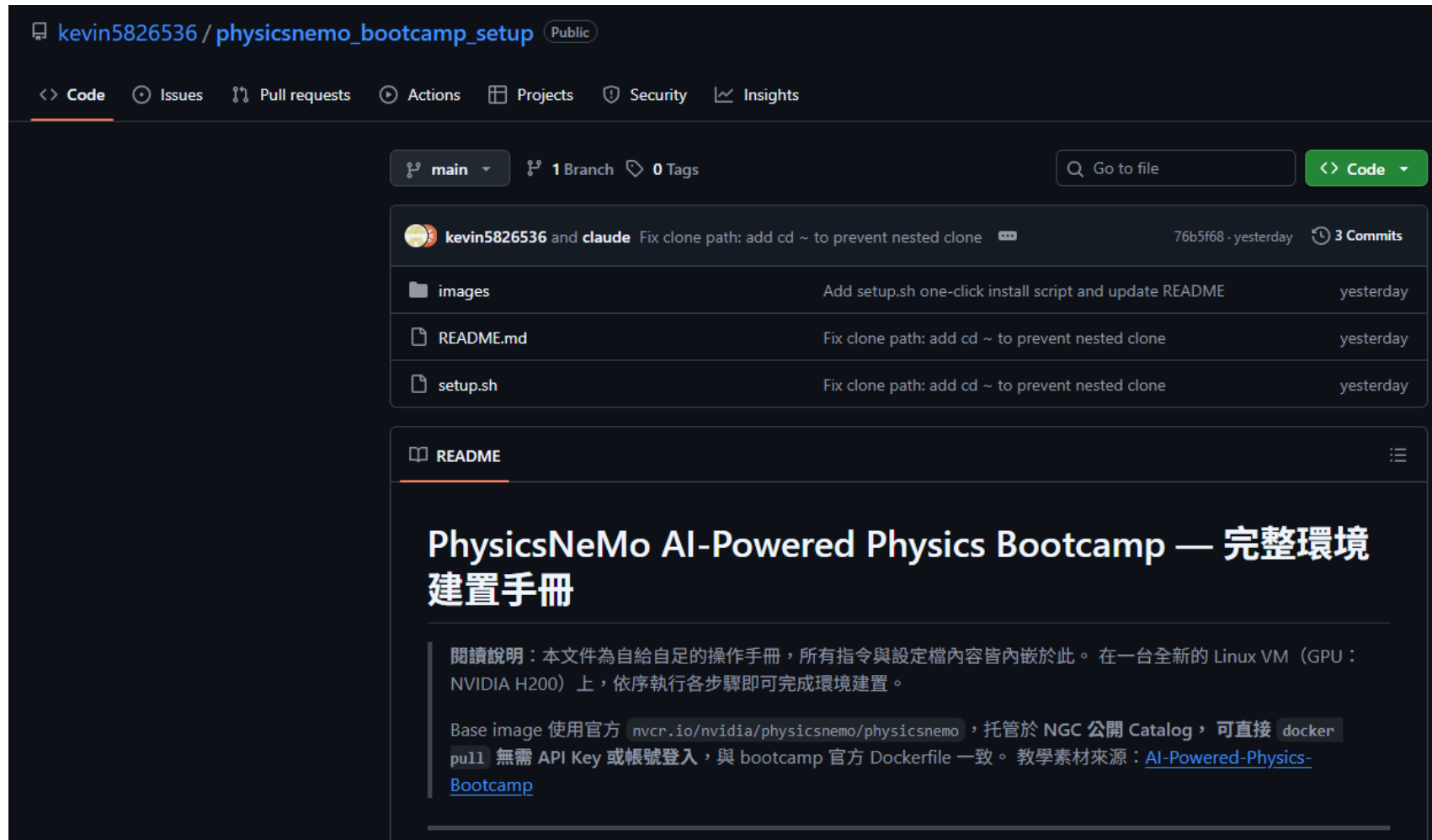
Hands-on with PhysicsNeMo

— Challenges

Bootcamp Dev Setup Instructions, see short URL below

<https://shorturl.at/Xq4fv>

(from long url: https://github.com/kevin5826536/physicsnemo_bootcamp_setup)



kevin5826536 / physicsnemo_bootcamp_setup Public

<> Code Issues Pull requests Actions Projects Security Insights

main 1 Branch 0 Tags Go to file <> Code

kevin5826536 and claude Fix clone path: add cd ~ to prevent nested clone 76b5f68 · yesterday 3 Commits

images	Add setup.sh one-click install script and update README	yesterday
README.md	Fix clone path: add cd ~ to prevent nested clone	yesterday
setup.sh	Fix clone path: add cd ~ to prevent nested clone	yesterday

README

PhysicsNeMo AI-Powered Physics Bootcamp — 完整環境建置手冊

閱讀說明：本文件為自給自足的操作手冊，所有指令與設定檔內容皆內嵌於此。在一台全新的 Linux VM（GPU：NVIDIA H200）上，依序執行各步驟即可完成環境建置。

Base image 使用官方 `nvcr.io/nvidia/physicsnemo/physicsnemo`，托管於 NGC 公開 Catalog，可直接 `docker pull` 無需 API Key 或帳號登入，與 bootcamp 官方 Dockerfile 一致。教學素材來源：[AI-Powered-Physics-Bootcamp](#)

Challenges

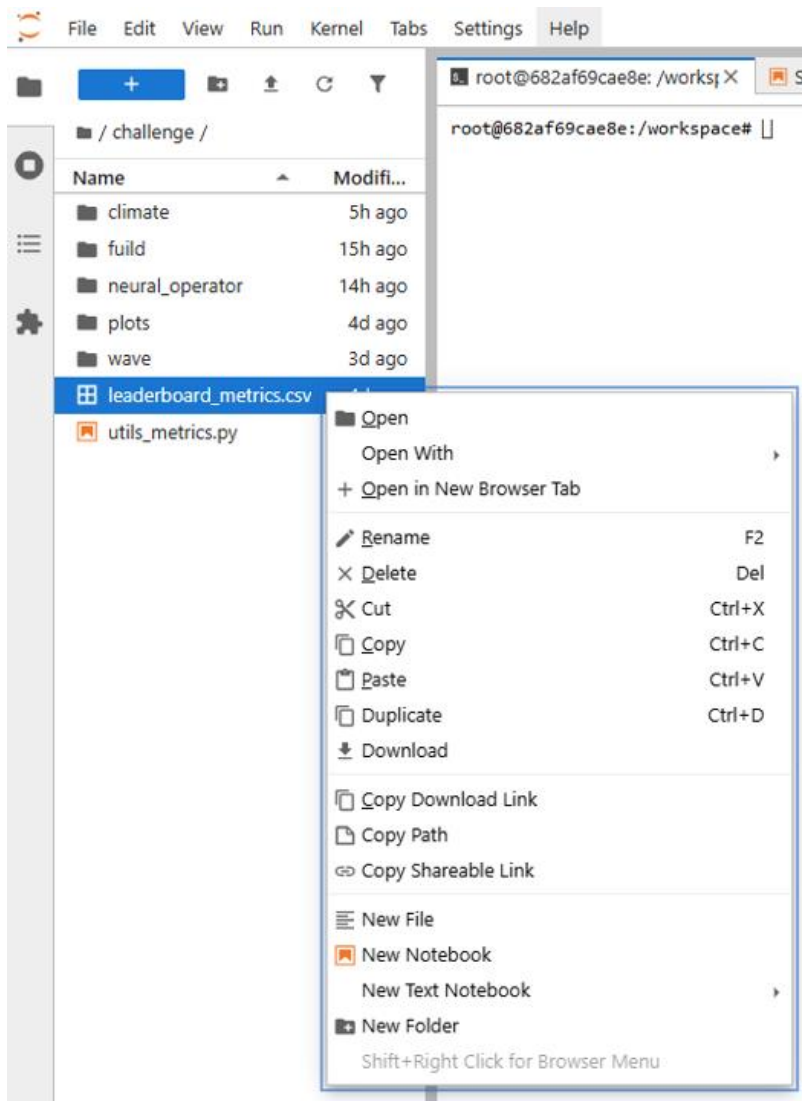
We have seen some hands-on examples on Physics informed approach, let us now use PhysicsNeMo to solve challenges.

1. Advanced Wave Dynamics
 - Level 1 Basic 2D wave equation
 - Level 2 Add variable wave speed
 - Level 3 Multiple sources and complex boundaries
2. Fluid-Structure Interaction
 - Level 1 2D flow
 - Level 2 Multiple Blocks in Channel
 - Level 3 Time-Dependent Flow
3. Multi-Physics Climate Modeling
 - Level 1 Simple Atmosphere Model
 - Level 2 Coupled Atmosphere–Ocean System
4. Advanced Neural Operators
 - Level 1 Fourier Neural Operator
 - Level 2 Adaptive Fourier Neural Operator
 - Level 3 Physics-Informed Neural Operator

Leaderboard

<https://shorturl.at/NVVQV>

(from long url: <https://drive.google.com/drive/folders/1nZRAMdVwUUBYQBVAuLsWDdEX4-GQan-x>)



上傳至記分板：

1. 將下載的 `leaderboard_metrics.csv` 壓縮成 `.zip` 檔，檔名格式為 `xxx_1.zip`（`xxx` 可自訂，例如團隊名稱）
2. 上傳至：[記分板 Google Drive 資料夾](#)

⚠ 注意事項

- CSV 檔名必須完全相同：zip 壓縮包內的檔案名稱必須是 `leaderboard_metrics.csv`，一字不差。若檔名有任何變動（例如重新命名），背景表單將無法解析，分數欄位會顯示為 `NA`。
- 每次重新提交必須遞增編號：若已上傳過 `xxx_1.zip`，再次上傳同名 `xxx_1.zip` 不會更新記分板。請改用 `xxx_2.zip`、`xxx_3.zip`（依序 +1）重新上傳，`xxx` 部分需與第一次相同。



Reveal of Answers and Final Rankings

Rankings



THANK YOU